

# Data Logging System for Physiological Parameters

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# Introduction

- ▶ Monitoring physiological parameters is an important part of modern healthcare systems.
- ▶ Continuous monitoring helps doctors detect abnormal health conditions at an early stage.
- ▶ Physiological parameters such as Heart Rate, Blood Pressure, Oxygen Saturation and Temperature indicate the health status of a patient.
- ▶ Traditional monitoring systems require manual observation by medical staff.
- ▶ Automated monitoring systems can continuously track patient data and improve response time.
- ▶ Micro-controller based systems allow efficient collection and processing of physiological data.
- ▶ Data logging systems help store and analyze patient health information over time.
- ▶ Alert systems can notify medical staff when abnormal readings occur.



# Problem Statement

- ▶ Continuous monitoring of patient physiological parameters is critical in healthcare environments.
- ▶ In many cases, patient parameters are monitored manually, which can be time consuming and prone to human error.
- ▶ Sudden abnormal changes in physiological parameters may not be detected immediately.
- ▶ Lack of automated alert systems can delay medical response during critical conditions.
- ▶ There is a need for a system that can simulate, monitor and analyze physiological data continuously.
- ▶ The system should also generate alerts when abnormal parameter values are detected.



# Objectives

- ▶ To simulate physiological parameter data using the TM4C123GXL microcontroller.
- ▶ To collect and use the simulated data as input for the monitoring system.
- ▶ To analyze and visualize physiological parameters using graphical representation.
- ▶ To monitor parameters such as Heart Rate, SpO2, Respiratory Rate, Temperature and Blood Pressure.
- ▶ To detect abnormal parameter values based on predefined normal ranges.
- ▶ To implement an alert generation system for abnormal readings.
- ▶ To broadcast alerts to systems running the receiver program.



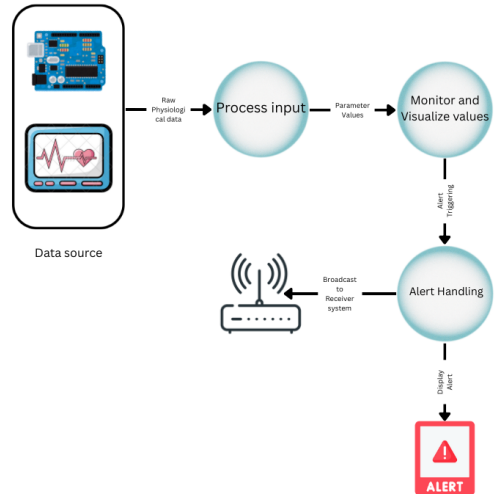
# System Overview

- ▶ The system simulates physiological parameter data using the TM4C123GXL microcontroller.
- ▶ The generated data represents patient health parameters such as Heart Rate, SpO2, Respiratory Rate, Temperature and Blood Pressure.
- ▶ This simulated data is transmitted to a monitoring program.
- ▶ The monitoring system processes the incoming data for analysis.
- ▶ Graphs are plotted to visualize trends of each physiological parameter.
- ▶ The system checks whether the values fall within normal medical ranges.
- ▶ If abnormal values are detected, an alert message is generated.
- ▶ The alert is broadcasted to all systems running the receiver program.



# System Workflow

- ▶ Data Simulation using TM4C123GXL
- ▶ Transmission of physiological data
- ▶ Data processing and monitoring
- ▶ Graphical visualization of parameters
- ▶ Detection of abnormal values
- ▶ Alert generation
- ▶ Broadcast alerts to receiver systems



# Physiological Parameters Monitored

- ▶ Heart Rate (HR)
- ▶ Oxygen Saturation (SpO2)
- ▶ Respiratory Rate (RR)
- ▶ Body Temperature
- ▶ Blood Pressure (BP)

|    | Timestamp      | PatientID | HR  | BP      | SpO2 | Temp | RR |
|----|----------------|-----------|-----|---------|------|------|----|
| 1  | 20260311192813 | P12345    | 118 | 171/106 | 83   | 38.5 | 27 |
| 2  | 20260311192816 | P12345    | 94  | 110/80  | 99   | 36.4 | 12 |
| 3  | 20260311192819 | P12345    | 75  | 117/83  | 99   | 36.7 | 16 |
| 4  | 20260311192822 | P12345    | 86  | 123/75  | 96   | 36.9 | 12 |
| 5  | 20260311192825 | P12345    | 92  | 116/74  | 98   | 36.7 | 15 |
| 6  | 20260311192828 | P12345    | 86  | 110/74  | 100  | 37.0 | 16 |
| 7  | 20260311192831 | P12345    | 65  | 170/69  | 82   | 36.6 | 21 |
| 8  | 20260311192834 | P12345    | 81  | 124/73  | 99   | 36.9 | 16 |
| 9  | 20260311192837 | P12345    | 84  | 114/74  | 96   | 36.6 | 15 |
| 10 | 20260311192840 | P12345    | 78  | 121/74  | 96   | 36.4 | 13 |
| 11 | 20260311192843 | P12345    | 67  | 122/77  | 100  | 36.7 | 12 |
| 12 | 20260311192846 | P12345    | 82  | 112/83  | 100  | 36.7 | 14 |
| 13 | 20260311192849 | P12345    | 69  | 112/83  | 99   | 37.0 | 18 |
| 14 | 20260311192852 | P12345    | 86  | 111/78  | 96   | 36.5 | 16 |
| 15 | 20260311192855 | P12345    | 65  | 114/80  | 99   | 36.8 | 13 |
| 16 | 20260311192858 | P12345    | 70  | 125/76  | 97   | 36.4 | 14 |
| 17 | 20260311192901 | P12345    | 59  | 111/80  | 91   | 39.0 | 23 |
| 18 | 20260311192904 | P12345    | 72  | 124/84  | 99   | 36.7 | 14 |
| 19 | 20260311192907 | P12345    | 83  | 112/78  | 98   | 37.1 | 12 |





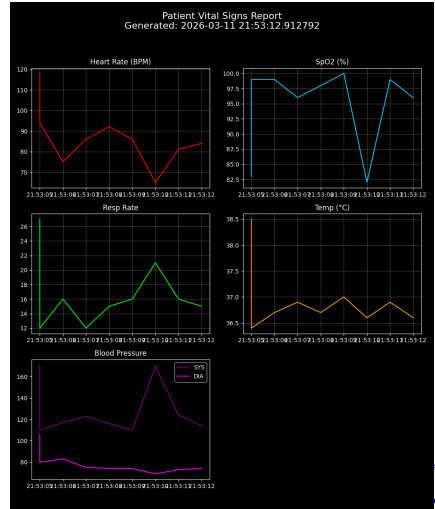
# Physiological Data Simulation

- ▶ The TM4C123GXL micro-controller is used to generate simulated physiological data.
- ▶ Random number generation techniques are used to create realistic parameter values.
- ▶ The generated data mimics real patient physiological signals.
- ▶ Each parameter is generated within medically accepted ranges.
- ▶ The simulated data is transmitted to the monitoring system for further processing.
- ▶ This approach allows testing of the system without requiring actual medical devices.



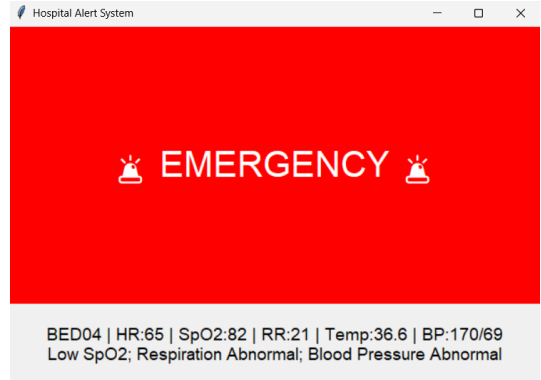
# Graph Plotting

- ▶ Physiological data is processed and stored.
- ▶ Graphs are plotted for each physiological parameter.
- ▶ Visualization helps understand parameter trends over time.
- ▶ Real-time graph updates allow continuous monitoring of patient health.
- ▶ Graphs help identify sudden changes or abnormal patterns in the data.
- ▶ Abnormal conditions can be easily detected through visual analysis.



# Alert Generation System

- ▶ The system continuously monitors incoming data.
- ▶ Each parameter is compared with predefined normal ranges.
- ▶ If abnormal values are detected, alerts are generated.
- ▶ Alerts are broadcasted to all systems running the receiver program.



# Work Completed in Phase 1

- ▶ Simulated physiological data using TM4C123GXL microcontroller.
- ▶ Used simulated data as input to the monitoring system.
- ▶ Plotted graphs for individual parameters.
- ▶ Implemented alert detection system.
- ▶ Broadcasted alerts to receiver systems.



- ▶ Integrate real physiological sensors with micro-controller.
- ▶ Implement real-time physiological data logging.
- ▶ Develop advanced centralized monitoring dashboard.
- ▶ Implement the connection with USB, Bluetooth and WiFi
- ▶ Improve alert notification mechanisms.



Many Thanks

